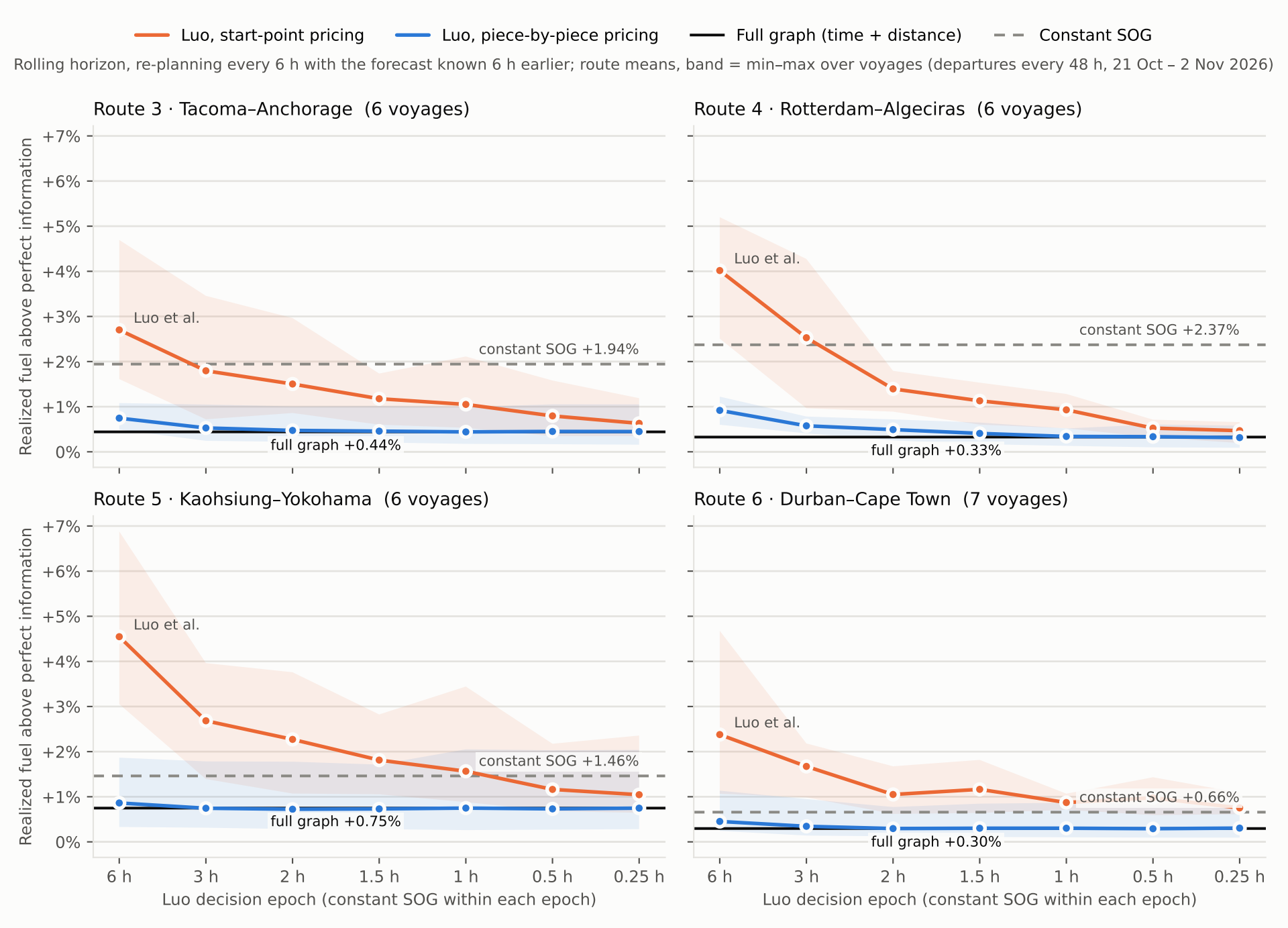
Luo, start-point pricing
Luo, piece-by-piece pricing
Full graph (time + distance)
Constant SOG
Rolling horizon, re-planning every 6 h with the forecast known 6 h earlier; route means, band = min–max over voyages (departures every 48 h, 21 Oct – 2 Nov 2026)
Route 3 · Tacoma–Anchorage (6 voyages)
Route 4 · Rotterdam–Algeciras (6 voyages)
Route 5 · Kaohsiung–Yokohama (6 voyages)
Route 6 · Durban–Cape Town (7 voyages)
Realized fuel above perfect information
+7%
+6%
+5%
+4%
+3%
+2%
+1%
0%
Luo et al.
constant SOG +1.94%
full graph +0.44%
constant SOG +2.37%
full graph +0.33%
constant SOG +1.46%
full graph +0.75%
constant SOG +0.66%
full graph +0.30%
6 h
3 h
2 h
1.5 h
1 h
0.5 h
0.25 h
Luo decision epoch (constant SOG within each epoch)